

Frequency or Information? A Controlled Study of Spectral Side-Channel Conditioning in Graph Diffusion

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Abstract

Spectral conditioning is standard in graph generation, but in existing generators the truncated spectrum *is* the representation being generated, which ties “which frequency band helps” to “how much of the graph that representation already carries.” We separate the two: we diffuse the full adjacency matrix and supply the truncated normalized-Laplacian spectrum only as a *side channel*, so band and cutoff k are the only quantities that vary, and we add a model-free probe measuring how much of the target each condition gives away. On SPECTRE’s Planar benchmark, five conditioning arms over five cutoffs show no frequency gradient: 23 of 25 cells sit within one standard deviation of the unconditioned baseline, and the effect is confined to two isolated minima that the probe separates. Low-frequency conditioning at $k=8$ improves the MMD Ratio $2.3\times$ while recovering 0.0% of the target’s edges; high-frequency conditioning at $k=32$ improves it $29.7\times$ but recovers 98.5%, a near-oracle regime rather than a frequency effect.

1 Introduction

Topology-only graph generation learns a distribution over adjacency matrices with no node or edge features, so connectivity is the only thing modelled. Diffusion models now dominate the area (Niu et al., 2020; Jo et al., 2022; Vignac et al., 2023), and a recurring idea is to hand the generator the graph’s Laplacian spectrum. Martinkus et al. (2022) generate eigenpairs with a GAN and decode an adjacency matrix from them; Minello et al. (2025) diffuse the spectrum itself; Vignac et al. (2023) feed Laplacian eigenfeatures to their denoiser as fixed extra inputs; and Zhu et al. (2025) argue, for representation learning, that the low-frequency part of the spectrum carries the useful global information.

These motivate an obvious question — *which* part of the spectrum helps generation? — that ex-

isting designs cannot answer cleanly. When the truncated spectrum (λ_k, U_k) is the representation the model generates and decodes from, k controls the frequency content of the condition *and* how much of the target that representation determines. On $n=64$ planar graphs $U_k \text{diag}(\lambda_k) U_k^\top$ at $k=32$ is close to the Laplacian itself, so a band that “wins” there may simply be inverting its own condition. No cited work reports a condition-information floor against which its frequency curve could be read.

Research question. *Given a diffusion model whose state already carries the full graph, does an explicit spectral side channel improve generation, and does the incremental benefit depend on which frequency band and how many eigenpairs are supplied?*

Phrasing it as a side channel makes the question falsifiable. Every arm diffuses the same object with the same architecture and budget and receives a condition tensor of the same shape; only the *selection* of eigenpairs differs. A none arm gives the null, and three controls — a random band, shape-matched Gaussian noise, and another graph’s real low band — separate frequency-specific information from conditioning capacity and from generic spectral statistics. We then add the measurement the literature lacks: a model-free probe that reconstructs edges from the condition alone.¹

Our contributions are that design; the central negative result that there is **no frequency gradient** (§4.2); a leakage probe that **separates the two minima** (§4.3), showing the study’s strongest number to be a near-oracle regime rather than a frequency effect; a diagnosis of the universal 0%

¹Code and run artifacts: https://github.com/TamaraBluzer/Frequency_aware_diffusion-scripts/make_paper_figures.py rebuilds both figures from results/*.json with no hand-typed number.

validity (§4.4); and two further negative results (§§4.5–4.6).

2 Related Work

SPECTRE (Martinkus et al., 2022) is the closest methodological and evaluation reference: a GAN generates (λ_k, U_k) on a Stiefel manifold, then a graph network builds the adjacency matrix from $U_k \text{diag}(\lambda_k) U_k^\top$. It sweeps $k \in \{2, 4, 8, 16, 32\}$ on Planar and SBM — the grid we reuse — but picks one k per dataset heuristically and never reports the curve. We treat k and the band as the object of study, replace its sign-canonicalization heuristic with a sign- and basis-invariant pair channel, and diffuse adjacency rather than decoding a generated spectrum.

GGSD (Minello et al., 2025) already measures smallest- versus largest-eigenpair representations inside a diffusion generator, so frequency-band sensitivity is not itself novel. Ours is narrower: the incremental value of a side channel while the diffusion state still carries the full graph, with matched-capacity controls and a measured information floor.

Others. **DiGress** (Vignac et al., 2023) supplies our datasets, discrete process and evaluation suite, and feeds Laplacian eigenvalues to its denoiser as a fixed augmentation rather than a variable to ablate — exactly the variable we isolate. **LGD** (Zhou et al., 2024) runs latent diffusion on molecules under scalar property conditioning; §3 says why we abandoned its design. **DualDiff** (Xie and Pan, 2026) couples node- and cluster-level latents, but a cluster count collapses global structure into one number, where an eigenpair-indexed condition exposes an ordered band and so admits a sweep. **SDMG** (Zhu et al., 2025) finds low-frequency reconstruction beating full-spectrum reconstruction for *representation* learning; we ask whether that transfers to *generation*.

3 Method

Data. SPECTRE’s released Planar benchmark: 200 graphs at $n=64$, each a Delaunay triangulation of uniform random points, split 128/32/40 following DiGress’s recipe, which we re-process ourselves since its `process()` appends every graph twice. Each arm generates 32 graphs, matching the evaluation set. **Every number here is scored on the validation split**, which also selected the

checkpoints (since lost), so all results are optimistic in the usual way.

Spectral condition. Let $L = I - D^{-1/2} A D^{-1/2} = U \Lambda U^\top$ with $0 = \lambda_1 \leq \dots \leq \lambda_n \leq 2$, and let t be the number of zero eigenvalues (equivalently, connected components). The arms take k eigenpairs as `low` = $(\lambda_{t+1} \dots \lambda_{t+k})$, `high` = $(\lambda_{n-k+1} \dots \lambda_n)$, `random` = k indices drawn uniformly from $t+1 \dots n$, `gaussian` = i.i.d. $\mathcal{N}(0, 1)$ tensors of identical shape, `shuffled` = another graph’s real low band under a derangement, and `none` = an all-zero condition with a presence flag.

The condition enters the denoiser twice: as a pair channel $(U_k \text{diag}(\lambda_k) U_k^\top)_{ij}$, rescaled per graph to unit RMS so `low` and `high` cannot differ in activation scale, and as the selected eigenvalues mapped from $[0, 2]$ to $[-1, 1]$, supplied globally with the timestep. The pair channel is invariant to eigenvector sign flips and to rotations within an exactly repeated eigenspace, sidestepping the ambiguity that motivates SignNet (Lim et al., 2023) without depending on it.

Model. The diffusion state is the centred adjacency matrix $x_0 = 2A - 1$. A cosine forward process (Nichol and Dhariwal, 2021) adds symmetric Gaussian noise to each undirected pair, and a permutation-equivariant DiGress transformer predicts x_0 from x_t (Ho et al., 2020; Vignac et al., 2023), masking the diagonal and padding at every step. It has 4 layers ($d_x=128$, $d_e=64$, $d_y=128$, 8 heads; 1.83M parameters), $T=100$ steps, and trains 200 epochs at batch 16, learning rate 3×10^{-4} , with the condition dropped at probability 0.1 to enable guidance at sampling time (Ho and Salimans, 2022). We began from an LGD-style autoencoder and abandoned it after measuring it: its per-pair latent is an expansion, not a compression (32,768 floats for 2,016 binary decisions), and reaches F1 = 1.000 only by copying A_{ij} down a private lane.

Evaluation. We reimplement the GraphRNN/GRAN/SPECTRE suite (You et al., 2018; Liao et al., 2019; Martinkus et al., 2022): MMD (Gretton et al., 2012) under a total-variation Gaussian kernel over degree histograms, clustering coefficients, normalized-Laplacian eigenvalues, 12 ab-spline wavelet scales, and 4-node orbit counts via ORCA (Hočevár and Demšar, 2014). Raw MMD is uninterpretable,

so we report **Ratio**: each metric divided by its train-vs-test floor and averaged, so 1.0 means indistinguishable from real data at this sample size. Validity is `networkx` planarity *and* connectivity (Hagberg et al., 2008). We calibrated the harness before training any model of ours: real-vs-real scores ≈ 0 , the floor is 1.00 by construction, and a *per-graph density-matched* Erdős–Rényi control (Erdős and Rényi, 1959) scores $3035\times$ it.

Protocol and leakage probe. Each arm uses three seeds at equal compute; band comparisons use a paired hierarchical bootstrap (1,000 resamples) over graphs and seeds. The **leakage probe** is model-free: it scores every node pair by $(U_k \text{diag}(\lambda_k) U_k^\top)_{ij}$ — exactly the tensor the denoiser receives — and keeps the m highest-scoring pairs, where m is the target’s true edge count. Using the true edge and node counts is deliberately privileged: the probe upper-bounds what any decoder could extract, so a low score is strong evidence a band cannot leak. Gaussian conditions score at chance, and the full spectrum recovers 100.0%, as it must.

4 Results

Three run families. Results below come from three run sets at identical architecture and budget: the $k=8$ *pilot* (Table 1), the *sweep* (Figure 2, seed 0 only), and the *seed confirmation* of the two minima (Table 2). Each therefore estimates the same unconditioned baseline — 327.2, 335.7, 330.1, agreeing to within one seed standard deviation — and `low` $k=8$ appears as 158.0, 116.3 and 141.9, a spread consistent with that arm’s own ± 19.3 across seeds. Every comparison below is *within* one set, against that set’s own baseline.

4.1 Conditioning helps at $k=8$

Table 1 is the kill gate we set before anything downstream. Low-frequency conditioning halves the Ratio and cuts validation loss by 60%, while high and random conditioning are indistinguishable from `none`. Paired bootstrap 95% intervals for `low` minus `none`, `high` and `random` are $[-189.4, -152.1]$, $[-206.7, -152.4]$ and $[-195.1, -142.2]$, all excluding zero by a wide margin. The model demonstrably uses the condition: under identical diffusion noise, adding it changes 12.95% of edge decisions. Figure 1 shows the gap. All arms match the target density and `none` is planar in any of its 32 samples,

band	Ratio ↓	val. loss ↓	valid
none	327.2 ± 8.8	0.132 ± 0.012	0%
low	158.0 ± 16.2	0.053 ± 0.004	0%
high	339.1 ± 9.2	0.094 ± 0.009	0%
random	326.5 ± 11.8	0.128 ± 0.012	0%

Table 1: Oracle conditioning at $k=8$, three seeds, mean $\pm$ std. Validation loss is metric-free and orders the arms the same way Ratio does. This is the $k=8$ pilot, a separate run family from the sweep of §4.2; §4 reconciles the two.

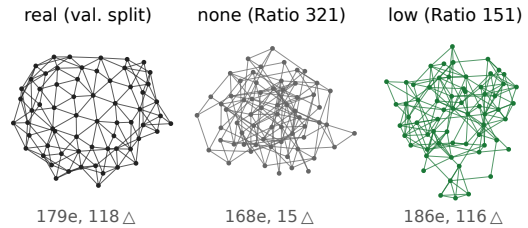


Figure 1: Seed-0 samples of two Table 1 arms beside a real Planar graph, with edge and triangle counts. Index 0 of every arm, not a chosen one, under one shared Kamada–Kawai layout, so the real graph is *not* handed its triangulation’s coordinates. `high` and `random` look like `none` and are omitted.

so validity cannot separate them; triangle count can. Against a real graph’s 118, `low` manages 116 where `none`, `high` and `random` manage 15, 15 and 13: what the low band supplies is the local mesh a triangulation is made of.

4.2 The sweep: two spikes, not a gradient

SDMG’s intuition implies quality should vary smoothly with how much of which end of the spectrum is supplied. It does not (Figure 2(a)). Over the 23 conditioned cells that are not one of the two minima, mean Ratio is 327.1 ± 18.5 , and the unconditioned baseline of 335.7 sits $+0.46$ standard deviations *inside* that spread: the flat region simply is the baseline. Each of those cells is a single seed, so we claim only that they lie within one standard deviation of the baseline, not that each is individually tested. Against that spread the minima sit -10.0 (`low`, $k=8$, at its three-seed mean) and -17.1 (`high`, $k=32$) standard deviations below, while their neighbours in the same band show nothing — on the same seed-0 sweep `low` is 316.4 at $k=4$ and 270.5 at $k=16$, against 116.3 at $k=8$.

Both minima reproduce across three seeds (Table 2): `high`, $k=32$ is very stable (std 0.58 on a mean of 11.11) and the only arm in the sweep

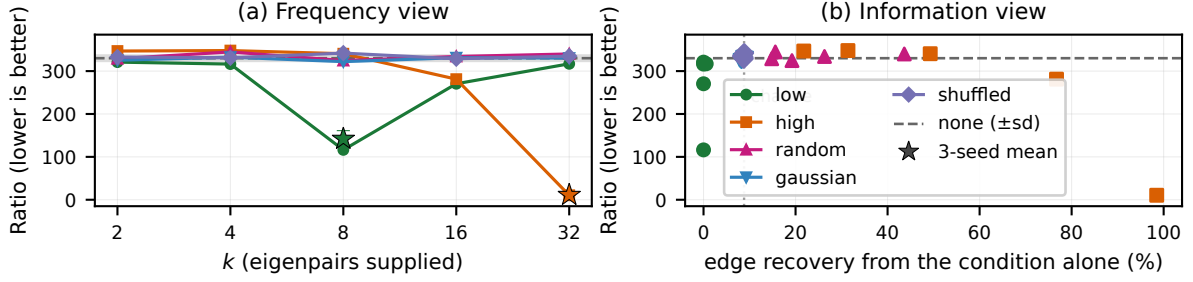


Figure 2: Five conditioning arms $\times$ five cutoffs on Planar, seed 0; stars mark three-seed means $\pm$ std for the two re-run cells. **(a)** Quality against k , baseline dashed. **(b)** The same points against measured leakage.

arm	Ratio $\downarrow$	vs. base	leak
high $k=32$	11.1 ± 0.6	$29.7\times$	98.5%
low $k=8$	141.9 ± 19.3	$2.3\times$	0.0%
none	330.1 ± 6.2	—	—

Table 2: Three-seed confirmation of the two minima, with edge recovery from the condition alone, against this family’s own *none* baseline. Both reproduce; only one of them is leak-free.

band	$k=2$	$k=4$	$k=8$	$k=16$	$k=32$
low	0.0	0.0	0.0	0.0	0.6
high	21.8	31.4	49.3	76.7	98.5
random	14.8	15.6	19.2	26.3	43.6
gaussian	7.9	8.5	8.2	8.7	9.4
shuffled	8.7	8.7	8.8	8.7	8.3

Table 3: Edge recovery (%) from the condition alone, Planar validation split; chance is 8.8%. The probe uses the true edge count, so these are upper bounds.

that produced a valid planar graph. *low*, $k=8$ is noisier but never approaches baseline, and its seed-0 draw was its most favourable, so 141.9 is the number to quote. The controls do their job: *gaussian* spans 322–333 and *shuffled* — another graph’s real low band — spans 329–342, both flat on the baseline at every k . Neither *having a condition* nor *having real spectral statistics* suffices; the effect needs the matching graph’s own spectrum at a particular band and cutoff, the cleanest evidence we have that the mechanism is informational rather than extra capacity.

4.3 Leakage separates the two minima

An obvious objection to Table 1: each sample is conditioned on a validation graph’s spectrum and scored against those same graphs, so a band that hands over more of the target wins for reasons unrelated to frequency. Table 3 measures this, and at $k=8$ the objection is refuted in the strongest pos-

sible way: leakage is *anti*-correlated with quality. *low* recovers 0.0% of edges — below chance — and wins by ~ 170 Ratio points, while *high* recovers 49.3% and loses. The pattern replicates on SBM (*low* at or below 3.5% against 9.4% chance, *high* 87.7% at $k=32$), so it is not an artifact of Planar’s construction from 2D points.

At $k=32$, however, leakage and quality *coincide* (Figure 2(b)): the minima sit at opposite ends of the information axis and are not the same phenomenon. *low*, $k=8$ improves generation while leaking nothing measurable, so it must supply coarse global structure the model cannot otherwise infer — the one genuine frequency effect here. *high*, $k=32$ supplies a near-complete specification of the target, so it is closer to handing the model the answer than to selecting a frequency. Reporting its $29.7\times$ improvement without its 98.5% leakage would be misleading — exactly the confound the information floor was added to catch.

4.4 Why validity is zero everywhere

Validity is *connected AND planar*, so the universal 0% cannot say which half failed; the probe’s reconstructions separate them. Connectivity is easy: by $k=32$ every reconstruction is connected in both bands, and *low* at $k=16$ is already 71.9% connected on 0.0% edge recovery. Planarity essentially never is — only *high*, $k=32$ reaches it, at 3.1%.

That arm explains why. At 98.5% recovery the mean number of misplaced edges is 2.7 of ~ 177 (max 6), and the fraction with *zero* misplaced edges is 3.1% — the same 3.1% that is planar, so every reconstruction with even one wrong edge is non-planar. This is a property of the target class, not our model: a planar graph on $n=64$ with 177 edges sits near the $3n - 6 = 186$ bound, so it

is nearly maximally triangulated and almost any added chord creates a K_5 or $K_{3,3}$ minor. Validity is therefore close to a binary test of exactness, not a graded measure, which is why an arm can improve Ratio $2.3\times$ while pinned at zero. The usual prescription — add structural features — has negative precedent in our own runs: the discrete arm of §4.5 includes cycle counts and still reports 0% validity at every scale.

4.5 From oracle to end-to-end

Everything above is oracle-conditioned: the spectra come from held-out real graphs, so these arms are a diagnostic upper bound rather than a generative model. To close it we trained a diffusion prior over (n, λ_k, U_k) with an orthogonality penalty and a QR retraction at the final step, then sampled end to end. Against an ER reference of 161.67, oracle spectra give Ratio 5.92 (high, $k=32$) and 74.86 (low, $k=8$); end to end at guidance scale 1 these become 159.72 and 295.64, so the low arm fails its gate outright. Guidance is U-shaped over $\{0.5, 1, 1.5, 2, 3\}$, optimal at 1.5, and even there a $24\times$ degradation remains, because small eigenvector errors compound through $U \text{diag}(\lambda) U^\top$. These Ratios omit ORCA and are *not* comparable to the tables above, hence the ER reference.

A discrete-diffusion arm (Austin et al., 2021) sharpens the caveat. At a serious budget (10 layers, 500 steps, 1,000 epochs) the *unconditioned* baseline improves $\sim 8\times$, from ~ 327 to 39.60; conditioning still helps, to 15.40 (-61%), but the gap narrows from ~ 220 Ratio points to ~ 24 . A small-budget headline overstates conditioning’s contribution, and validity stays 0% at every scale.

4.6 Downstream utility: a negative result

The intended payoff is synthetic augmentation. We test it on a task deliberately aligned with the hypothesis — predicting an SBM graph’s community count (2–5), which is what low-frequency eigenvalues encode — with 20/40/80 real training graphs and 5 seeds, each augmented graph conditioned on a real training graph and inheriting its label. Diffusion augmentation never beats real-only training: at $n=20$ unconditioned augmentation hurts (89.8% against 94.8%) and conditioned augmentation is no better (92.0%); at $n=80$ both stay below the 97.2% baseline. Fresh samples from the *true* SBM generator do help slightly (98.8% at $n=80$), locating the failure in sample quality rather than the pipeline. The task is also

easy to begin with, so this shows the gains of §4.2 do not automatically become downstream utility, not that they never could.

5 Discussion and Conclusion

What the side channel buys. Very little, almost everywhere: given a diffusion state that already carries the full graph, a truncated spectrum is worthless in 23 of 25 configurations, and the two exceptions differ in kind. The real finding is `low, k=8` — a $2.3\times$ Ratio improvement and 60% lower validation loss from a condition a privileged probe cannot use to recover one edge above chance, hard to explain as anything but coarse global structure that is useful and hard to infer alone. `high, k=32` is the better number and the weaker result: without the probe we would have reported a near-oracle regime as our headline. That a controlled study can invert its own top-line claim is our best argument for an information floor none of the cited work reports.

Two caveats on our own reporting. First, the sharpness is unexplained: both minima reproduce ten to seventeen standard deviations below a flat surface, so they are real, but no mechanism predicts *which* cells spike. Second, Ratio averages whichever metrics were available, and orbit’s tiny floor dominates any mean it enters: `seed-0 none` is 321.21 with ORCA and 125.25 without, a $2.56\times$ gap that brackets the unexplained $2.09\times$ ER reference shift (337.91 vs. 161.67). Orbit contributes 84.0% of the $k=8$ improvement while *degree moves the wrong way*, and a geometric mean puts the aggregate at -34.8% rather than -52.9% . The low-band result survives — it wins on four of five metrics and on validation loss — but the arithmetic mean overstates it.

Limitations. (i) Every number is validation-split, on splits that also selected checkpoints, and validity is 0% throughout, so we compare distributional fidelity, not sample correctness. (ii) The quality sweep is Planar only — on SBM we ran the leakage half, which replicates, not the generation half — and SBM validity uses a spectral-clustering stand-in. (iii) The sweep’s values come from a recycled Colab stdout, so its per-metric breakdowns are lost; ORCA was verified present, so they are five-metric. (iv) Our discrete model is not a DiGress reproduction, and four planned ablations were not run.

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